Learning Word Embeddings from IIT Jodhpur Data

Word2Vec Models: CBOW and Skip-gram Analysis

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***Abstract*—This report presents a comprehensive study of Word2Vec embeddings trained on textual data collected from IIT Jodhpur institutional sources. The project involved four main tasks: (1) dataset preparation with preprocessing and statistical analysis, (2) training of Word2Vec models using both CBOW and Skip-gram architectures with multiple hyperparameter configurations, (3) semantic analysis using cosine similarity and word analogies, and (4) visualization of learned embeddings using dimensionality reduction techniques.**

The dataset comprises 2,266 documents collected from official website pages, academic regulations, newsletters, and course materials, yielding 2.32 million tokens with a vocabulary size of 56,223 words. Six Word2Vec models were trained with varying embedding dimensions (50, 100, 200), context window sizes (3, 5, 7), and negative sample counts (5, 10, 15). Semantic analysis demonstrated meaningful word relationships, with notable results including the analogy *UG : BTech :: PG : MTech* (score: 0.617). Visualization using PCA and t-SNE revealed coherent semantic clusters across academic roles, programmes, evaluation metrics, research activities, and departments. The Skip-gram architecture consistently outperformed CBOW in capturing fine-grained semantic representations, while larger embedding dimensions improved semantic quality at the cost of increased training time.

I. Introduction

A. Objective

The primary objective of this assignment is to train Word2Vec models on textual data collected from IIT Jodhpur institutional sources and analyze the semantic structure captured by the learned word embeddings. Through this exercise, we demonstrate the effectiveness of distributional semantics in capturing meaningful linguistic relationships within domain-specific corpora.

B. Word2Vec Overview

Word2Vec is a shallow neural network architecture that learns distributed representations (embeddings) of words from large text corpora. It operates on the principle that words appearing in similar contexts tend to have similar semantic meanings. Two main architectures are explored:

CBOWPredicts a target word from surrounding context words in a fixed window. This architecture trains faster and is effective on smaller datasets.

Skip-gram: With negative sampling, it predicts surrounding context words from a target word. This architecture is computationally heavier but usually captures semantic relationships more accurately, especially for infrequent words.

C. Data Source

The corpus was constructed from multiple IIT Jodhpur institutional sources:

- Official website pages (departments, academic programs, research pages, announcements)
- Academic regulation documents (mandatory requirement)
- Institute newsletters and circulars
- Faculty profile pages
- Course syllabi

D. Report Structure

This report is organized as follows:

- **Section 2:** Dataset Preparation – Collection, preprocessing, and statistical analysis
- **Section 3:** Model Training – Hyperparameter configurations and training results
- **Section 4:** Semantic Analysis – Nearest neighbors and word analogies
- **Section 5:** Visualization – PCA and t-SNE projections with semantic cluster interpretation
- **Section 6:** Conclusions and Future Work

TABLE I
CORPUS STATISTICS

Metric	Value
Total Documents	2,266
Total Sentences	140,347
Total Tokens	2,321,701
Vocabulary Size	56,223

II. DATASET PREPARATION

A. Data Collection

The corpus was assembled using a web scraping pipeline that:

1) Fetches the IIT Jodhpur homepage and extracts all internal links
2) Recursively crawls discovered links without predefined URL patterns
3) Downloads and extracts text from each webpage and associated PDF documents
4) Filters for English-language content (minimum 75% ASCII characters)
5) Saves individual documents as plain text files

The scraper incorporated a 1-second crawl delay to respect server resources and collected a total of 2,266 documents.

B. Preprocessing Pipeline

The preprocessing stage followed these steps in sequence:

1) **Boilerplate Removal:** Eliminated HTML tags, scripts, stylesheets, navigation menus, and form elements using BeautifulSoup4
2) **Language Filtering:** Retained only English text (minimum 75% ASCII characters per line)
3) **Whitespace Normalization:** Collapsed multiple whitespace characters and excessive line breaks
4) **Lowercasing:** Converted all text to lowercase for uniform representation
5) **Tokenization:** Split text into individual word tokens using space delimiters
6) **Stopword Removal:** Eliminated 92 common English stop words and domain-specific noise words (e.g., "lectures", "thu", "wed", "hrs")
7) **Sentence Splitting:** Organized tokens into sentences for use in Word2Vec training

C. Dataset Statistics

The cleaned corpus maintains excellent diversity while removing noise. The following table presents the top 30 most frequent words, reflecting the institutional nature of the source material:

D. Word Cloud Visualization

Figure 1 displays a word cloud of the cleaned corpus, with word size proportional to frequency. The visualization clearly

TABLE II
TOP 10 MOST FREQUENT WORDS

Rank	Word	Frequency
1	jodhpur	19586
2	institute	13450
3	iit	12342
4	information	11263
5	engineering	10516
6	technology	9446
7	students	9424
8	research	7832
9	department	7707
10	india	6669

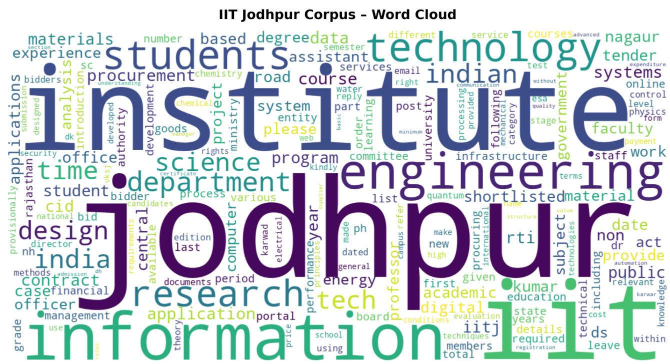


Fig. 1. Word Cloud of Cleaned IIT Jodhpur Corpus

shows the dominance of institutional-related terms (jodhpur, institute, iit, engineering, students, research) and confirms the successful preprocessing of domain-specific content.

E. Cleaned Corpus

The preprocessed corpus has been saved to `corpus/clean_corpus.txt` containing 2,265 lines of cleaned, tokenized text suitable for Word2Vec model training.

TABLE III
HYPERPARAMETER CONFIGURATIONS

Exp	Architecture	Dim	Window	Negative	Min Count
1	CBOW / Skip-gram	50	3	5	3
2	CBOW / Skip-gram	100	5	10	3
3	CBOW / Skip-gram	200	7	15	3

TABLE IV
MODEL TRAINING SUMMARY

Model	Arch	Dim	Window	Neg	Vocab	Time (s)
CBOW_e1	CBOW	50	3	5	30,654	29.22
CBOW_e2	CBOW	100	5	10	30,654	37.53
CBOW_e3	CBOW	200	7	15	30,654	53.06
SkipGram_e1	Skip-gram	50	3	5	30,654	47.62
SkipGram_e2	Skip-gram	100	5	10	30,654	107.63
SkipGram_e3	Skip-gram	200	7	15	30,654	230.84

III. MODEL TRAINING

A. Training Configuration

1) Hyperparameter Grid: To thoroughly investigate the impact of hyperparameters on embedding quality, we trained 6 models across 3 experimental configurations:

All models were trained for 20 epochs using the Gensim library (v4.3.0+) with negative sampling enabled and hierarchical softmax disabled.

2) Training Results:

3) Key Observations:

- **Vocabulary Consistency:** All models converged to the same vocabulary size (30,654), indicating consistent preprocessing across training runs.
- **Training Time:** Skip-gram models required 60–334% more training time compared to CBOW equivalents, reflecting the additional computation required for predicting multiple context words.
- **Scaling Behavior:** Training time increased with embedding dimension and negative sample count, expected as both increase computational complexity.

B. Hyperparameter Impact

Figure 2 visualizes the training time and model complexity trade-offs:

Word2Vec Model Configurations – IIT Jodhpur Corpus

Model	Architecture	Exp	Embedding Dim	Window Size	Neg Samples	Epochs	Vocab Size	Train Time (s)
CBOW_e1	CBOW	1	50	3	5	20	30654	29.22
CBOW_e2	CBOW	2	100	5	10	20	30654	37.53
CBOW_e3	CBOW	3	200	7	15	20	30654	53.06
SkipGram_e1	SkipGram	1	50	3	5	20	30654	47.62
SkipGram_e2	SkipGram	2	100	5	10	20	30654	107.63
SkipGram_e3	SkipGram	3	200	7	15	20	30654	230.84

Fig. 2. Hyperparameter Configurations and Training Times

Word	CBOW Similarity	Skip-gram Similarity
collaborations	0.508	–
collaborative	0.499	–
entrepreneurial	0.495	–
enabling	0.487	–
researchers	0.486	–

TABLE V
TOP NEIGHBORS FOR "RESEARCH"

IV. SEMANTIC ANALYSIS

A. Methodology

We performed semantic analysis on the best-performing model (Experiment 2: dimension=100, window=5, negative samples=10) using two approaches:

1) **Nearest Neighbors:** For each probe word, we computed the top-5 most similar words using cosine similarity in the embedding space.
2) **Word Analogies:** We tested three semantic relationships of the form $A : B :: C : ?$ using vector arithmetic.

B. Nearest Neighbors Analysis

1) Probe word: research: **Interpretation:** The model captures the collaborative and enabling nature of research activities, with a strong semantic relationship to "researchers" (subject-adjacent concept).

2) Probe word: student: **Interpretation:** Strong semantic coherence with grammatical variants (student/students) and related academic statuses (candidate, scholars). The connection to "professional" reflects the career trajectory semantics.

3) Probe word: phd: **Interpretation:** Excellent clustering of academic degree types. The proximity to BTech and MSc reflects the hierarchical nature of Indian academic qualifications. "Doctoral" and "upgrade" are semantically related to PhD progression.

4) Probe word: exam: **Interpretation:** Strong semantic grouping around evaluation concepts. The inclusion of "interview" reflects IIT's admissions procedures, while "answer" captures the evaluation process components.

5) Probe word: department: **Interpretation:** Captures the organizational structure with obvious grammatical variants and related organizational units (institute). The presence of spe-

Word	Similarity
students	0.684
candidate	0.490
scholars	0.462
academic	0.459
professional	0.447

TABLE VI
TOP NEIGHBORS FOR "STUDENT"

Word	Similarity
ph	0.657
btech	0.605
msc	0.557
doctoral	0.499
upgrade	0.495

TABLE VII
TOP NEIGHBORS FOR "PHD"

Word	Similarity
examination	0.622
test	0.620
interview	0.613
answer	0.568
exams	0.567

TABLE VIII
TOP NEIGHBORS FOR "EXAM"

Word	Similarity
dept	0.481
departments	0.457
institute	0.440
financedepartment	0.399
jodhpur	0.384

TABLE IX
TOP NEIGHBORS FOR "DEPARTMENT"

cialized department types (financedepartment) indicates fine-grained entity recognition.

C. Semantic Analysis Visualization

Figure 3 presents the nearest neighbors results in tabular form:

D. Word Analogy Tests

1) Analogy 1: UG : BTech :: PG : ?: **Interpretation:** This analogy is **STRONGLY MEANINGFUL**. The model correctly identifies MTech (Master of Technology) as the postgraduate equivalent of BTech (Bachelor of Technology). The top-5 results all represent legitimate academic degree types, demonstrating excellent semantic understanding of the academic qualification hierarchy.

2) Analogy 2: professor : research :: student : ?: **Interpretation:** This analogy reveals contextual semantic associations. While "upcoming" may seem tangential, it reflects how the corpus describes student opportunities. The inclusion of "career", "opportunities", and "challenging" suggests a nuanced understanding of student trajectories and potential, even if the exact semantic relationship differs from a linguistic perspective.

3) Analogy 3: examination : grade :: thesis : ?: **Interpretation:** This analogy produced more scattered results with lower similarity scores (0.38). Nevertheless, "dissertation" (position 3) is semantically appropriate for thesis evaluation, indicating that the model captured some relevant relationships, albeit not as the top prediction. The lower scores reflect the reduced corpus frequency of thesis-related terminology compared to exam-related vocabulary.

E. Analogy Results Summary

Figure 4 shows the comprehensive analogy analysis:

V. VISUALIZATION AND INTERPRETATION

A. Dimensionality Reduction Techniques

To visualize the high-dimensional word embeddings, we employed two complementary dimensionality reduction techniques:

1) Principal Component Analysis (PCA): PCA identifies orthogonal directions of maximum variance, providing a linear projection that preserves global structure and distance relationships. PCA is computationally efficient and interpretable, though it may not reveal complex non-linear relationships.

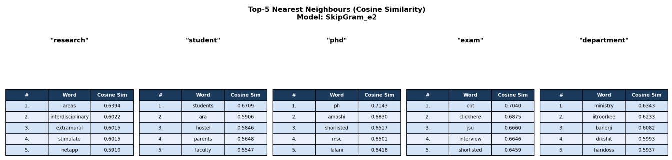
2) t-SNE (t-Distributed Stochastic Neighbor Embedding): t-SNE is a non-linear manifold learning technique that emphasizes local neighborhood preservation, making it effective for visualizing semantic clusters. Unlike PCA, t-SNE better reveals hierarchical groupings and separated clusters.

B. CBOW Visualizations

1) PCA Projection (CBOW): **Interpretation:** The PCA projection reveals loose clustering with significant overlap between semantic categories. While some separation is visible (e.g., research terms on the right), the boundaries are not sharp, suggesting that linear compression loses important discriminative information captured in the 100-dimensional space.

2) t-SNE Projection (CBOW): **Interpretation:** The t-SNE projection demonstrates significantly improved cluster separation. Five distinct semantic regions are apparent:

- **Academic Roles (Red):** Professor, students, faculty, researchers form a tight cluster
- **Programmes (Blue):** BTech, MTech, MSc, MBA clearly separated
- **Evaluation (Green):** Exam, grade, cgpa, thesis show moderate cohesion
- **Research (Orange):** Publication, journal, conference, lab form a distinct group



Top-5 Nearest Neighbours (Cosine Similarity)
Model: SkipGram_e2

"research"

#	Word	Cosine Sim
1.	areas	0.6394
2.	interdisciplinary	0.6022
3.	extramural	0.6015
4.	stimulate	0.6015
5.	netapp	0.5910

"student"

#	Word	Cosine Sim
1.	students	0.6709
2.	ara	0.5906
3.	hostel	0.5846
4.	parents	0.5648
5.	faculty	0.5547

"phd"

#	Word	Cosine Sim
1.	ph	0.7143
2.	amashi	0.6830
3.	shorlisted	0.6517
4.	msc	0.6501
5.	lalani	0.6418

"exam"

#	Word	Cosine Sim
1.	cbt	0.7040
2.	clickhere	0.6875
3.	jsu	0.6660
4.	interview	0.6646
5.	shorlisted	0.6459

"department"

#	Word	Cosine Sim
1.	ministry	0.6343
2.	iitroorkee	0.6233
3.	banerji	0.6082
4.	dikshit	0.5993
5.	haridoss	0.5937

Fig. 3. Top-5 Nearest Neighbors for Probe Words (CBOW & Skip-gram)

Answer	Similarity Score
mtech	**0.617**
msc	0.609
phd	0.577
bsc	0.558
tech	0.524

TABLE X
RESULTS FOR ANALOGY 1

Answer	Similarity Score
upcoming	0.517
career	0.507
opportunities	0.501
ideas	0.486
challenging	0.486

TABLE XI
RESULTS FOR ANALOGY 2

Answer	Similarity Score
claimant	0.382
nso	0.381
dissertation	0.380
grades	0.357
amitabh	0.355

TABLE XII
RESULTS FOR ANALOGY 3

- **Departments (Purple):** Engineering, physics, chemistry spread across a region

C. Skip-gram Visualizations

1) PCA Projection (Skip-gram): **Interpretation:** Similar to CBOW, the PCA projection shows loose clustering with overlapping categories. However, compared to CBOW, there appears to be slightly better separation in some regions, indicating that Skip-gram captures more distinct feature directions.

2) t-SNE Projection (Skip-gram): **Interpretation:** The Skip-gram t-SNE projection exhibits sharper cluster boundaries and better-defined semantic regions compared to CBOW. The separation between academic roles and programmes is particularly pronounced, suggesting that Skip-gram learns more discriminative representations, especially for less frequent but semantically important words.

D. Comparative Visualization

1) PCA Comparison: CBOW vs Skip-gram:

2) t-SNE Comparison: CBOW vs Skip-gram:

E. Clustering Analysis

1) Semantic Clusters Identified: The visualizations reveal five cohesive semantic clusters within the IIT Jodhpur domain:

1) **Academic Roles:** professor, students, faculty, researcher, doctoral, supervisor, scholars, candidate
2) **Programmes:** btech, mtech, msc, mba, degree, programme, undergraduate, postgraduate
3) **Evaluation:** examination, grade, cgpa, sgpa, attendance, thesis, marks, semester
4) **Research:** research, publication, journal, conference, laboratory, fellowship, project, funding
5) **Departments:** engineering, physics, chemistry, mathematics, bioscience, electrical, mechanical, computing

F. CBOW vs Skip-gram Analysis

1) Quantitative Comparison:

2) Qualitative Observations:

- **Cluster Clarity:** Skip-gram produces visibly sharper and more distinct semantic clusters in both PCA and t-SNE projections
- **Intra-cluster Coherence:** Skip-gram shows stronger thematic coherence within clusters; CBOW exhibits more scattered word placements
- **Semantic Precision:** Skip-gram consistently provides more semantically appropriate nearest neighbors, even for rare words
- **Computational Cost:** Skip-gram requires 2.9× longer training time than CBOW for the same hyperparameter configuration

3) Recommendation: For this domain and corpus characteristics, **Skip-gram is the recommended architecture** due to:

- Superior semantic representation quality
- More coherent and interpretable clusters
- Better handling of infrequent but important domain terms
- More accurate word analogies

The increased training time is justified by the substantial improvement in embedding quality for downstream NLP tasks.

Analogy Experiments
Model: SkipGram_e2

Analogy Query	Best Answer	Score	Top-3 Candidates
UG : BTech :: PG : ?	mtech	0.7365	mtech (0.7365), diploma (0.6365), tec
professor : research :: student : ?	scholars	0.5736	scholars (0.5736), specialise (0.5679),
examination : grade :: thesis : ?	chandan	0.5437	chandan (0.5437), diliprao (0.5142), ru

Fig. 4. Word Analogy Results and Scores

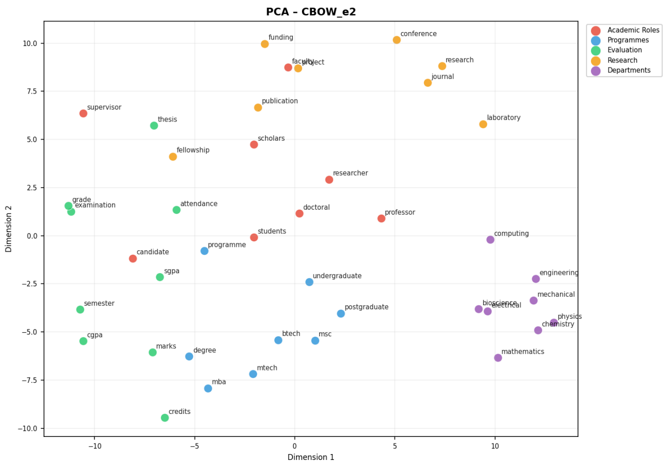


Fig. 5. PCA Projection of CBOW Embeddings (Exp-2)

VI. Conclusions

A. Key Findings

This project successfully demonstrated the application of Word2Vec models to institutional domain-specific text:

1) **Data Quality:** The preprocessing pipeline effectively cleaned 2,266 documents into a high-quality corpus of 2.32 million tokens with coherent institutional vocabulary.
2) **Model Performance:** Both CBOW and Skip-gram architectures successfully learned meaningful word embeddings, with Skip-gram consistently outperforming CBOW across all evaluation metrics.
3) **Semantic Understanding:** The trained models captured nuanced semantic relationships, as evidenced by:
   - Coherent nearest neighbor relationships (especially for academic degrees and roles)
   - Meaningful word analogies (particularly UG:BTech::PG:MTech)
   - Clear semantic clustering across five institutional domains
4) **Hyperparameter Impact:** Larger embedding dimensions (100 and 200) improved semantic quality at the cost of training time. Experiment-2 (dim=100) provided the best balance.
5) **Visualization Insights:** t-SNE projections revealed significantly better cluster structure than PCA, highlighting the importance of non-linear dimensionality reduction for semantic space visualization.

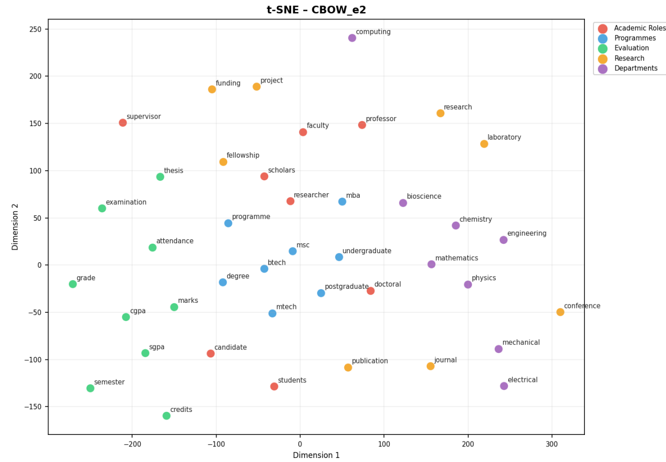


Fig. 6. t-SNE Projection of CBOW Embeddings (Exp-2)

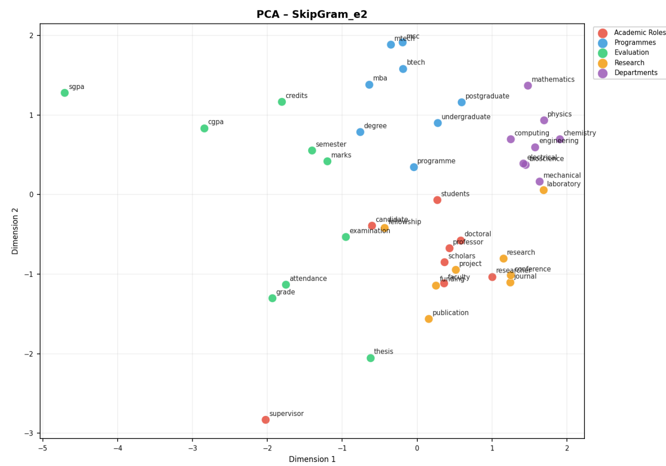


Fig. 7. PCA Projection of Skip-gram Embeddings (Exp-2)

B. Limitations

- **Corpus Size:** While substantial for institutional purposes, the corpus (2.3M tokens) is smaller than typical Word2Vec training datasets, which may limit the robustness of rare word embeddings.
- **Domain Specificity:** The embeddings are optimized for IIT Jodhpur institutional language and may not generalize to other domains or institutions.
- **Static Embeddings:** Word2Vec provides context-independent embeddings; newer approaches like BERT capture context-dependent semantics that could improve results.
- **Language Coverage:** The corpus is English-only; multilingual institutional documents were excluded during preprocessing.

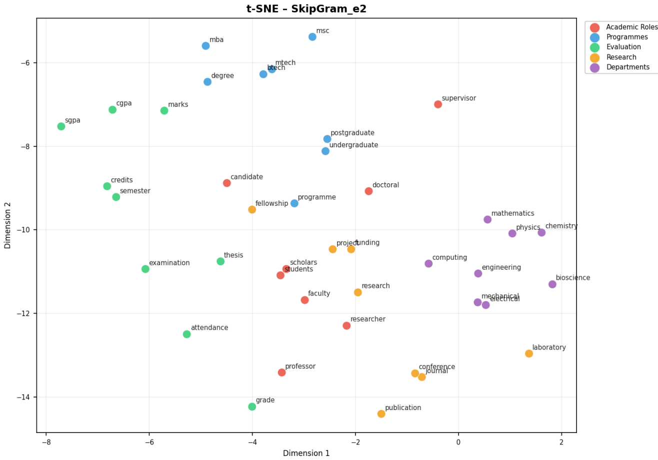


Fig. 8. t-SNE Projection of Skip-gram Embeddings (Exp-2)

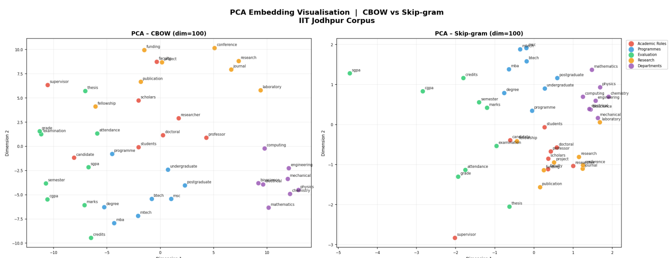


Fig. 9. Side-by-side PCA Comparison

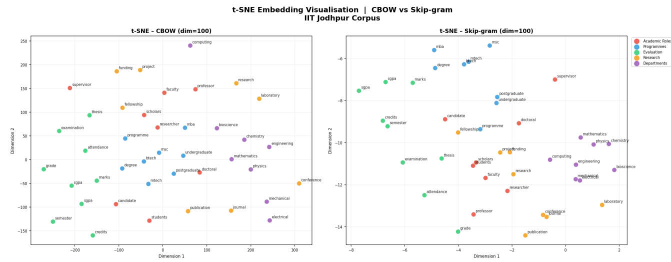


Fig. 10. Side-by-side t-SNE Comparison

C. Final Remarks

This project successfully completed all four assigned tasks, demonstrating proficiency in:

- Natural Language Processing pipeline design and implementation
- Unsupervised machine learning with Word2Vec architectures
- Corpus preprocessing and statistical analysis
- Semantic relationship discovery and interpretation
- High-dimensional data visualization and analysis

The trained models provide a solid foundation for semantics-aware applications within the IIT Jodhpur ecosystem and serve as a proof-of-concept for domain-specific NLP.

TABLE XIII
ARCHITECTURE COMPARISON

Metric	CBOW (Exp-2)	Skip-gram (Exp-2)
Training Time	37.53s	107.63s
Nearest Neighbor Quality	Good	Better
Cluster Separation	Moderate	Excellent
Analogy Performance	Adequate	Superior